

Fernando d'Ávila Lins Bezerra Cavalcanti Filho

AI Software Engineer

Building LLM Applications • AI Agents • Full Stack Systems

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Summary

AI-focused Software Engineer passionate about building intelligent products powered by Large Language Models. Experienced designing AI agents, Retrieval-Augmented Generation (RAG) applications, OCR pipelines, REST APIs and scalable web applications using Python, TypeScript, React, Next.js and PostgreSQL.

Strong builder mindset with experience taking products from idea to implementation through internships, AI research, university extension programs and startup environments. Passionate about rapidly learning new technologies, shipping production-ready software and solving real-world problems with AI.

Core Technologies

Artificial Intelligence

LLMs • AI Agents • RAG • Prompt Engineering • MCP • LangChain • LangGraph • NLP • OCR • PyTorch • Scikit-Learn • OpenAI APIs

Backend

Python • TypeScript • Node.js • REST APIs • PostgreSQL • SQL • Docker • Git • Linux

Frontend

React • Next.js • TailwindCSS • JavaScript • HTML • CSS

Practices

Software Architecture • Clean Code • Agile • API Design • System Integration • Problem Solving

Professional Experience

Full Stack Software Engineer Intern

Acutis Tecnologia

May 2024 – Present

- Engineered an OCR pipeline that automated invoice processing and significantly reduced manual data extraction.
- Designed and developed internal dashboards used for project management and operational visibility.
- Built REST APIs and backend services integrating frontend applications, databases and external services.
- Developed AI-powered learning tools including a flashcard platform focused on knowledge retention.
- Worked across the entire software lifecycle, including requirements gathering, implementation, testing, debugging and deployment.
- Collaborated in Agile teams using Git, Scrum and modern software engineering practices.

Software Engineering Intern

OpenDoor App

Jun 2025 – Present

- Developed Full Stack features for a home-swapping platform using React, Next.js, Node.js and PostgreSQL.
- Designed backend services, API endpoints and business workflows supporting the product architecture.
- Improved platform scalability through backend refactoring and database optimizations.
- Collaborated closely with product and engineering teams to rapidly ship new customer-facing features.

Jan 2024 – Jan 2025

- Built a scheduling optimization system that automated teacher allocation across multiple school classes.
- Replaced a manual scheduling workflow with a software solution developed in Python.
- Worked directly with stakeholders to transform operational requirements into production software.

Software Developer

Structuring Businesses Extension Program

Aug 2022 – Jul 2023

- Taught Lean Startup methodologies, MVP design and product validation alongside multidisciplinary university teams.
- Built a web platform for managing cattle competition events using TypeScript and JavaScript.
- Gathered business requirements and transformed them into software features through iterative development.

Selected AI Products

AI Subscription Auditor (Corta AI)

- Designed and built an AI-powered SaaS that analyzes bank statements to identify recurring software subscriptions, unused licenses and cost-saving opportunities for small and medium-sized businesses.
- Developed intelligent workflows capable of extracting financial insights and generating executive recommendations using Large Language Models.
- Built the product from ideation and architecture to frontend, backend and AI integration.
- **Stack:** Python, TypeScript, Next.js, PostgreSQL, OpenAI APIs.

MCP Security Scanner

- Built a security tool capable of detecting Tool Poisoning attacks against Model Context Protocol (MCP) servers.
- Implemented heuristic analysis for malicious tool descriptions and prompt injection patterns.
- Designed the project to demonstrate emerging security risks in AI agent ecosystems.
- **Stack:** Python, MCP, AI Agents.

AI Academic Assistant

- Built a Retrieval-Augmented Generation assistant capable of answering university-related questions using institutional documents.
- Implemented document ingestion, embeddings, semantic retrieval and context-aware answer generation.
- Designed the architecture to reduce hallucinations through grounded responses.
- **Stack:** Python, LangChain, OpenAI APIs, RAG.

Multiple Sclerosis MRI Detection

- Developing Machine Learning models for lesion detection and segmentation in brain MRI images.
- Working on medical image preprocessing, model evaluation and deep learning techniques.
- Research developed in collaboration with LABEN.
- **Stack:** Python, PyTorch, Scikit-Learn.

Interactive Quiz Simulator

- Built a concurrent Java application simulating multiplayer quiz sessions.
- Applied Object-Oriented Programming and concurrent programming concepts.

AI Leadership

TAIL — Artificial Intelligence Academic League

NLP Division

- Evaluated Large Language Models for Brazilian electronic invoice product normalization.
- Designed prompt engineering strategies and comparative evaluation methodologies.
- Developed AI prototypes involving NLP, sentiment analysis and LLM-powered applications.
- Collaborated with multidisciplinary teams researching practical applications of Generative AI.

Research Experience

LABEN — Aging and Neuroscience Research Laboratory

- Researcher working on Artificial Intelligence applied to medical imaging.
- Developing Machine Learning solutions for Multiple Sclerosis lesion detection.

Teaching

Teaching Assistant — Vector Calculus and Analytical Geometry

Federal University of Paraíba

- Mentored undergraduate students through weekly tutoring sessions.
- Received the **2023 Teaching Initiation Award (1st Place)** for excellence in academic mentoring.

Education

Federal University of Paraíba (UFPB)

Expected Graduation: 2027

Bachelor of Science in Computer Science

Relevant areas:

Artificial Intelligence • Machine Learning • Software Engineering • Algorithms • Data Structures • Databases • Computer Networks

Additional Experience

TRIL — Software Development Laboratory

- Participated in collaborative software engineering projects involving web applications and backend development.

Formula E-Motion

- Developed embedded systems and telemetry solutions for an electric Formula SAE race car.
- Worked with CAN Bus communication and electronic circuit development using KiCad.

Open Source & Continuous Learning

- Regularly explore new AI technologies including OpenAI, Anthropic, LangGraph, MCP, AWS Bedrock, Vector Databases and Agentic AI architectures.
- Build personal projects to evaluate new LLM capabilities, prompting techniques and AI frameworks.
- Continuously study recent AI research papers and software engineering best practices.

Languages

Portuguese (Native) — English (Professional Working Proficiency) — Spanish (Intermediate)

Interests

Artificial Intelligence • Agentic Systems • Large Language Models • Applied Machine Learning • Backend Engineering • Developer Tools • Startups • Product Engineering